



Erasify

Local AI-Powered Watermark & Logo Removal Platform

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Guide**

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1. Introduction

Erasify is a secure, local, high-fidelity platform engineered to remove watermark stamps and logo overlays from AI-generated media (such as Gemini and Veo style image/video outputs).

Unlike traditional inpainting models that "hallucinate" pixel fills and destroy surrounding details, Erasify operates using a mathematically exact **Reverse Alpha Blending** engine. It localizes watermark layers down to subpixel offsets, performs inverse blending, and removes the watermark transparently. This leaves the original background pixels untouched and maintains maximum quality.

Security Note: All media processing, decoding, and rendering take place strictly inside the user's local browser memory. Files are never uploaded to any external server, ensuring complete data privacy.

2. Server Requirements

To deploy and host the Erasify server backend, your server must meet the following minimum specs:

Requirement	Specification
Runtime Environment	Node.js >= 20.0.0 (LTS recommended)
Package Manager	pnpm >= 10.0.0 (or npm >= 10.0.0)
Database	MongoDB >= 6.0 (Atlas Cloud or Local Instance)
RAM	Minimum 1 GB (2 GB recommended for serverless or container runs)
Browser Support	Modern browsers (Chrome, Edge, Firefox, Safari) with WebAssembly and WebCodecs enabled

3. Installation

Follow these steps to set up the project locally:

- 1. Clone the project repository to your server:

```
git clone https://github.com/exobunix/erasify.git
cd gemini-watermark-remover
```

- 2. Install dependencies using `pnpm`:

```
pnpm install
```

- 3. Configure your environment variables by copying the template file:

```
cp .env.example .env
```

- 4. Run the production build script to compile the frontend assets to the `dist/` folder:

```
pnpm build
```

4. Environment Variables

The server reads configuration from a `.env` file in the project root. Below are the key environment variables:

Variable	Default	Description
<code>PORT</code>	<code>3000</code>	The port the Express server will listen on.
<code>MONGODB_URI</code>	<code>mongodb://localhost:27017/erasify</code>	The connection string to your MongoDB database.
<code>NODE_ENV</code>	<code>development</code>	Set to <code>production</code> in live environments to enable asset

5. Gemini API Configuration

Because Erasify operates as a sandboxed browser client (via Chrome Extension or Tampermonkey Userscript), it interacts with the **Gemini Web App** at <https://gemini.google.com/app> dynamically.

The userscript intercepts generated asset requests and matches original image bindings from Gemini's RPC calls (specifically targeting `hNvQHb` and `c8o8Fe` executions). This allows full-resolution downloads and clipboard copy processes to bypass watermarks on the client side without needing backend API keys.

6. Database Setup

The application uses MongoDB to manage user accounts, sessions, subscription tiers, and quota limits.

Atlas Cloud Network Whitelisting

If hosting the server on Vercel or any other serverless container platform, the IP addresses of the serverless function containers will change dynamically.

Crucial Step: You must configure Network Access in your MongoDB Atlas Dashboard to allow connections from anywhere: add `0.0.0.0/0` to the IP access list. Failure to do so will cause secure socket connections to time out.

Database Indexing

Upon initial boot, the application automatically ensures the following indexes are created on the `users` collection:

```
db.users.createIndex({ email: 1 }, { unique: true })
```

7. Running the Application

Commands to run and manage the process:

- **Development Mode** (Start local file watching and esbuild live reload):

```
pnpm dev
```

- **Production Start** (Launch the Node.js production server):

```
pnpm start
```

8. Admin & User Authentication

User Sign In & Registration

Users can register accounts via the frontend modal. Tiers include:

- **Free Tier:** Limits processing to 1 image, 0 videos.
- **Basic Tier:** Unlimited image removals, 0 videos.
- **Daily Tier:** 15 images/day, 3 videos/day.
- **Unlimited Tier:** Unlimited image removals, 100 video removals.

Admin Diagnostics

Admins can check database connectivity and server system health by querying the secure diagnostics endpoint:

```
GET /api/debug/db
```

This endpoint returns the server node version, masked MongoDB URI, and current database connectivity status.

9. Troubleshooting

Symptom	Cause	Resolution
Database connection failed: Socket timed out	MongoDB Atlas Firewall blocking Vercel IP.	Verify that IP 0.0.0.0/0 is added to Network Access in Atlas. If using Free Tier, verify the cluster is not paused.
npm error Cannot read properties of null (isDescendantOf)	Lockfile conflict inside Vite/npm install.	Enable pnpm install in Vercel settings under Build & Deployment.
Old "Logged In as" menu layout still visible	Browser caching static JS file.	Clear browser cache or use cache-busting script paths (e.g. erasify.js?v=1.0.31).

10. Support Contact

For technical support, custom feature development, or integration queries, contact Avdar Innovations:

- **Company:** Avdar Innovations Pvt Ltd
- **Support Line / WhatsApp:** +91 9967853364